Dejan Grubisic

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Experience

Axiom Math — Palo Alto, CA
Founding Engineer — Jun. 2025 – Jul. 2026

- Led a 4-engineer team delivering a large-scale autoformalization pipeline
- Engineered an AI system for code optimization, formally verified in Lean
- Helped build Axiom's first agentic system for automated theorem proving in Lean
- Developed a system for scalable data collection and contributed to model training

Meta - Modern Recommendation Systems — Menlo Park, CA
AI Research Scientist — Jul. 2024 – Jun. 2025

- Co-designed foundation models for retrieval, ranking, and content understanding
- Optimized Facebook Feed ranking model training for GPU efficiency and scalability

Meta - Fundamental AI Research (FAIR) — Menlo Park, CA
Machine Learning Research Intern — May 2023 – Dec. 2023

- Contributed to Meta's LLM Compiler, a suite of foundation models for compiler optimization
- Fine-tuned Llama 2-based models on LLVM IR programs to solve the phase-ordering problem
- Analyzed LLM capability in applying 60 LLVM optimizations

Meta - Fundamental AI Research (FAIR) — New York, NY
Machine Learning Research Intern — May 2022 – Dec. 2022

- Built LoopTune, a reinforcement learning framework for optimizing tensor operations
- Built and tuned cost and policy deep learning models

Rice University — Houston, TX
Graduate Research Assistant - PhD Research — Aug. 2019 – May 2024

- Developed GPU support for HPCToolkit, a large-scale performance profiler
- Created performance analysis techniques for GPU-accelerated applications
- Developed profiler-guided optimization based on reinforcement learning (Compiler2)

Berkeley Lab — Berkeley, CA
High Performance Computing Research Intern — Jun. 2021 – Sep. 2021

- Profiled and analyzed power consumption of multi-node GPU applications

Education

Doctor of Philosophy | ***High Performance Computing and Machine Learning*** — Aug. 2019 – May 2024
Rice University | GPA: 3.56 — Houston, TX

Master of Science | ***Big Data Architectures*** — Aug. 2018 – May 2019
University of Novi Sad | GPA: 4.00 — Novi Sad, Serbia

Bachelor of Science | ***Electrical and Computer Engineering*** — Aug. 2014 – May 2018
University of Novi Sad | GPA: 3.96 — Novi Sad, Serbia

Selected Publications

D. Grubisic et al. Compiler Generated Feedback for Large Language Models. *arXiv preprint*, 2024.
D. Grubisic et al. Priority Sampling of Large Language Models for Compilers. *EuroMLSys @ EuroSys*, 2024.
D. Grubisic et al. LoopTune: Optimizing Tensor Computations with Reinforcement Learning. *arXiv preprint*, 2023.
C. Cummins, V. Seeker, **D. Grubisic**, et al. Large Language Models for Compiler Optimization. *arXiv preprint*, 2023.
B. Wasti, **D. Grubisic**, et al. LoopStack: ML-friendly ML Compiler Stack. *ML for Systems Workshop @ NeurIPS*, 2022.
K. Zhou et al(incl **D. Grubisic**) An Automated Tool for Analysis and Tuning of GPU-Accelerated Code in HPC Application. *TPDS*, 2022.
K. Zhou et al(incl **D. Grubisic**) Measurement and Analysis of GPU-Accelerated Applications with HPCToolkit. *Parallel Computing*, 2021.

Selected Projects

Master Thesis — Multi-source shortest paths on dynamic graphs via Lambda architecture (pySpark, HDFS, Kafka, Docker)
Bachelor Thesis — FPGA core accelerating a chess engine; ESL design in SystemC, VHDL board evaluation, UVM verification
Parallel Computing — game-tree search (Cilk), LU decomposition (OpenMP), 2.5D matmul (OpenMPI), bitonic sort (CUDA)

Community and Leadership

Southeast Europe AI Summit — *Founder, connecting AI researchers, builders, and companies across Southeast Europe* — Jan. 2026 – Present

Technical Skills

Languages: Python, C/C++, CUDA C, Lean, Java, Bash, VHDL
Systems: PyTorch, LLVM, OpenMP, MPI, CUDA, Docker, Slurm, Spark, Linux
AI/ML: LLMs, fine-tuning, reinforcement learning, data preparation, compiler construction, parallel computing, formal verification